Reg. No. [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [18RADE721]

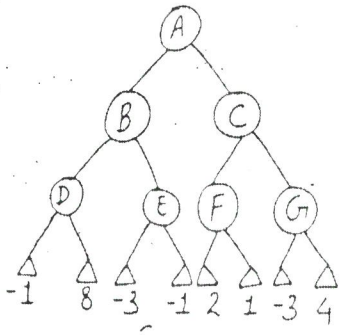
SEVENTH SEMESTER B.Tech EXAMINATION, DECEMBER–2021

ARTIFICIAL INTELLIGENCE FOR ROBOTICS

Time: 3 Hours **Maximum Marks: 70**

Instructions:
- **i.** Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 14 marks.
- **ii.** Missing data may be suitably assumed.

1. a. **07**



Describe the Min-Max search technique, using the above graph.

b. Describe a Simplex-reflex agent with a neat diagram. **07**

2. a. Explain the various edges in solving a plan graph problem with relevant illustrations/examples. **07**

b. Analyze the Partial-order Plan (PoP) problem, taking shoe-typing as an example. **07**

3. Write short notes on the following **14**

- **i.** Bayesian Networks
- **ii.** Making decision
- **iii.** Uncertainty

OR

4. a. A security guard at a underground center is able to find out if it has rained or not, observing everyday morning when the Director comes to office i.e. if s(he) is carrying an umbrella. In the above example. **09**

- **i.** What is Observable variable (E_t) and Unobservable (X_t)
- **ii.** Explain the 1st-order and 2nd order Markov process for the above problem with relevant illustrations.

b. Describe the probabilistic reasoning in Artificial Intelligence. **05**

5. a. Describe reinforcement learning with relevant examples. **07**

b. Explain the popular "Surprise candy problem" in the context of statistical learning techniques, assuming that there are two types of candies i.e. lime and cherry and like-wise make other relevant assumptions. **07**

OR

6. a. Describe Probabilistic language(speech recognition) processing using HMM. **07**

b. Briefly analyse the design of perception/vision from the perspective of an autonomous car. **07**

7. a. Explain the applications of Artificial Intelligence using suitable examples. **07**

b. Discuss the dynamics and control of robot movement, using PID controllers. **07**

OR

8. a. Briefly discuss motion planning in the context of **07**

i. Generalized configuration space and

ii. Recognizable sets

b. Analyze the statements "The success of AI might mean the end of the human race" and "People might lose their sense of being unique". **07**